Muqtadir Hussain

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Education

McMaster University September 2019 - April 2025
Bachelor of Engineering and Management (B.Eng.Mgt.), ***Engineering Physics*** *Hamilton, Ontario*

Skills

Programming & Development: Python, C/C++, JS, HTML/CSS, Git, Embedded Systems, PID Control

Modeling & Simulation: MATLAB, MCNP, ANSYS Fluent, TCAD (EDA), RSoft, Zemax, SPICE, Excel (VBA), Power BI

Instrumentation & Fabrication: CAD, 3D Printing, LIDAR/3D Metrology, FEA, GD&T, Cleanroom Microfabrication

Industrial: Technical Reporting, Root Cause Analysis, FMEA, Project Management, Stakeholder Liaison, R&D Support

Work Experience

Applied Precision Inc | Precision Metrology and Engineering Intern | Vaughan, ON Sept 2022 - Apr 2023

- Operated high-precision structured light and LIDAR scanners (Leica RTC 360, Zeiss Comet L3D, Faro) for reverse engineering and quality inspection across automotive, aerospace, and medical sectors.
- Processed and aligned point cloud data to generate watertight CAD models using Geomagic Design X and PolyWorks.
- Supported tooling inspections and component design workflows by preparing hybrid 3D/2D outputs for additive manufacturing and inspection reports.
- Contributed to technical research and marketing documentation while supporting logistics, sales, and documentation tasks.

Academic Projects

Autonomous Blackjack-Dealing System – Capstone Project Sep 2024 - Apr 2025

- Engineered and built a fully autonomous, gesture-controlled Blackjack dealer from concept to final demonstration
- Integrated a 3D-printed mechanical dispenser with an Arduino-controlled motor system that achieved an 85% single-card deal success rate
- The system's Python-based software used a MediaPipe neural network to classify 'Hit'/'Stand' hand gestures with 99.88% accuracy and a separate YOLOv8 model to recognize dealt cards, with all game states displayed through a custom Tkinter GUI.

SOTI Inc, Mississauga, ON – Industry Consulting Project Sep 2024 - Dec 2024

- Co-engineered an AI-driven autonomous client-engagement system to resolve lead-qualification and conversion inefficiencies for enterprise software firm SOTI, integrating business-process mapping with NLP-capable virtual agents.
- Built a data-backed cost & performance model (CAD $10k phased budget, KPIs, risk register); compared two architecture options with a weighted-scoring matrix and selected the higher-ROI solution for pilot rollout.
- Co-authored a client deck and executive presentation detailing system architecture, decision logic, risk mitigations and projected increases in leads and conversion (30% and 15% respectively), securing stakeholder approval

PERL Silicon Solar Cell Fabrication & Characterization Sep 2024 - Dec 2024

- Executed full-cycle fabrication (photolithography, doping, metallization) of a PERL silicon solar cell in a cleanroom environment , characterized performance under AM1.5G illumination (achieving 0.70% efficiency), and authored a scientific research paper detailing the methods, results, and analysis following McMaster Journal of Engineering Physics format

Monte Carlo Reactor Simulation – Fuel Bundle Optimization Jan 2024 - Apr 2024

- Modeled neutron transport and energy deposition in CANDU fuel bundles using MCNP; compared natural uranium, enriched uranium, MOX, and thorium fuels to optimize power output, flux uniformity, and thermal performance for core stability and efficiency; Delivered detailed technical recommendations in a formal presentation

McMaster Nuclear Reactor – Nuclear Laboratory Series Sep 2023 - Dec 2023

- Performed neutron activation, radiographic imaging, and attenuation experiments using HPGe, NaI(Tl), and BF_3/He-3 detectors; analyzed decay kinetics, cross-sections, and neutron-material interactions to validate shielding models and non-destructive inspection techniques.

PID-Controlled Thermoelectric Oven/Cooler Jan 2022 - Apr 2022

- Designed & implemented closed-loop PID temperature control on a TI MSP430 Microcontroller (in C), interfacing an NTC thermistor via ADC and modulating a Peltier TEC with PWM; built a MATLAB GUI for UART-based live plotting, setpoint input, and data logging